Home Assignment-3

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① min / max $f(x)$

sub to cons. $g_i(x) = 0$, $h_i(x) = 0$

S-1: Form Lagragian

$$L(x, \lambda, \mu) = f(x) + \sum \lambda_i g_i(x) + \sum \mu_i h_i(x)$$

S-2: stationary condition

$$\frac{\partial L}{\partial x} = 0$$

S-3: primal feasibility $g_i(g) \leq 0$, $h_j(x) = 0$

S-4: Dual feasibility $\lambda_i \geq 0$

S-5: Complementary stackness

$$\lambda_i g_i(x) = 0$$

S-6: solve all equations simultaneously.

② Given

Max $f = 4x_1 + 6x_2 - x_1^2 - 3x_2^2$

3 tc's $x_1 + 2x_2 \leq 4$, $x_1, x_2 \geq 0$

S-1: Convert to equality

$$x_1 + 2x_2 + s = 4$$

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S-2: $L = 4x_1 + 6x_2 - x_1^2 - 3x_2^2 + \lambda(4 - x_1 - 2x_2)$

S-3: partial derivative

$$\frac{\partial L}{\partial x_1} = 4 - 2x_1 - \lambda = 0 \quad -(1)$$

$$\frac{\partial L}{\partial x_2} = 6 - 6x_2 - 2\lambda = 0 \quad -(2)$$

S-4: solve

From (1): $\lambda = 4 - 2x$

sub in (2): $\rightarrow 6 - 6x_2 - 2(4 - 2x_1) = 0$

$\rightarrow 6 - 6x_2 - 8 + 4x_1 = 0$

$\rightarrow 4x_1 - 6x_2 = 2$

Constraints $x_1 + 2x_2 = 4$

S-5: solve system

from constraint, $x_1 = 4 - 2x_2$

$\rightarrow 4(4 - 2x_2) - 6x_2 = 2$

$\rightarrow 16 - 8x_2 - 6x_2 = 2$

$\rightarrow 16 - 14x_2 = 2$

$\boxed{x_2 = 1}$ $\boxed{x_1 = 2}$

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S-6: optimal value

$f = 8+6-4-3$ $\boxed{f = 7}$

$\boxed{\therefore (x_1, x_2) = (2,1), f_{max} = 7}$

③ Given

$\max\ z = 3x_1 + 4x_2$

$x_1^2 + x_2^2 \le 4$

S-1: $L = 3x_1 + 4x_2 + \lambda(4 - x_1^2 - x_2^2)$

S-2: Derivative, $3 - 2\lambda x_1 = 0$

$x_1 = \frac{3}{2\lambda}$

S-3: Substitute into constraints

$\left(\frac{3}{2\lambda}\right)^2 + \left(\frac{4}{2\lambda}\right)^2 = 4 \Rightarrow \frac{9+16}{4\lambda^2} = 4$

$\Rightarrow \frac{25}{4\lambda^2} = 4$ $\boxed{\lambda = \frac{5}{4}}$

S-4: values

$x_1 = \frac{3}{2(5/4)} = \frac{6}{5}$

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S-5: optimial value

$$z = \frac{18}{5} + \frac{32}{5} = 10$$

$$\boxed{\therefore (x_1, x_2) = \left(\frac{8}{5}, \frac{8}{5}\right), z_{max} = 10}$$

(4) Given

$$z = 2x_1^2 + x_2^2 + 3x_3^2 + 10x_1 + 8x_2 + 6x_3 + 100$$

$$x_1 + x_2 + x_3 = 20$$

S-1: $L = z + \lambda(20 - x_1 - x_2 - x_3)$

S-2: Derivatives

$$4x_1 + 10 - \lambda = 0$$

$$6x_3 - 6 - \lambda = 0$$

S-3: $x_4 = \frac{\lambda - 10}{4}, x_2 = \frac{\lambda - 8}{2}, x_3 = \frac{\lambda - 6}{6}$

Subs into Constraints

$$\frac{\lambda - 10}{4} + \frac{\lambda - 8}{2} + \frac{\lambda - 6}{6} = 20$$

solve $\rightarrow \lambda = 24$

S-4: values

$x_1 = 35, x_2 = 8, x_3 = 3$

$$\boxed{\therefore (x_1, x_2, x_3) = (35, 8, 3)}$$

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⑥ Min $z = 5x_1 + x_2^{-1} x_3^2 + x_1^{-2} x_2^{-1} + 10x_2^2 + 2x_1^{-1} x_2 x_3^{-2}$

$x_1, x_2, + x_3 = 0$ → This is a polynomial function

$z = T_1 + T_2 + T_3 + T_4$ where $T_1 = 5x_1 x_2^{-1} x_3^2$,

$T_2 = x_1^{-2} x_2^{-1}$, $T_3 = 10x_2^2$, $T_4 = 2x_1^{-1} x_2 x_3^{-2}$

S-2: Apply: GM inequality

$z \geq 4(T_1, T_2, T_3, T_4)^{1/4}$

S-3: $T_1 T_2 T_3 T_4 = (5)(1)(10)(2) x_1^{1-2-1} x_2^{-1+2+1} x_3$

$= 100\, x_1^{-2} x_2^{1} x_3^{0}$

$= 100 \frac{x_2}{x_2^1}$

S-4: for min $= T_1 = T_2 = T_3 = T_4$

from $T_1 = T_2$:

$5x_1 x_2^{-1} x_3^2 = x_1^{-2} x_2^{-1}$

from $T_2 = T_3$

$x_1^{-2} x_2^{-1} = 10x_2^2$

from ② $x_1^2 = \frac{1}{10x_2^3}$

sub into ③ & ④

$\boxed{x_1 = 1}$, $\boxed{x_2 = \frac{1}{2}}$, $\boxed{x_3 = 1}$

S-5: Min value

$z = 5(1)(2)(1) + (1)(2) + 10(1/4) + 2(1)(1/2)(1)$

$= 10 + 2 + 2.5 + 1$

$\boxed{z = 15.5}$

$\therefore x_1 = 1$

$x_2 = 1/2$

$x_3 = 1$

$z_{min} = 15.5$ //